

# **An Integrated Approach to Improving Child Health and Community Well-being**

**Securing Safe Water and Improving Sanitation,  
Strengthening Health and Livelihoods through Agroforestry,  
Reducing Child Labor, and Promoting CO<sub>2</sub> Sequestration**

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# **1. Introduction**

# 1-1. About This Proposal

This proposal takes an integrated approach to complex issues such as global **water** problems, poverty, **child labor**, and **CO<sub>2</sub>** emissions.

- With less than five years remaining until the 2030 SDG deadline, many challenges still remain.
- Many of these challenges are linked to the lack of safe, sanitary water.
- Increasing **CO<sub>2</sub>** sequestration can contribute to climate change mitigation and help reduce climate-related risks.

The factors behind the issues addressed by the SDGs are intricately interconnected.

## **Water & Sanitation**

- Global water issues
- Contaminated water
- Water collection (lost education)

## **Society & Economy**

- Poverty
- Child labor issues
- Unable to attend school

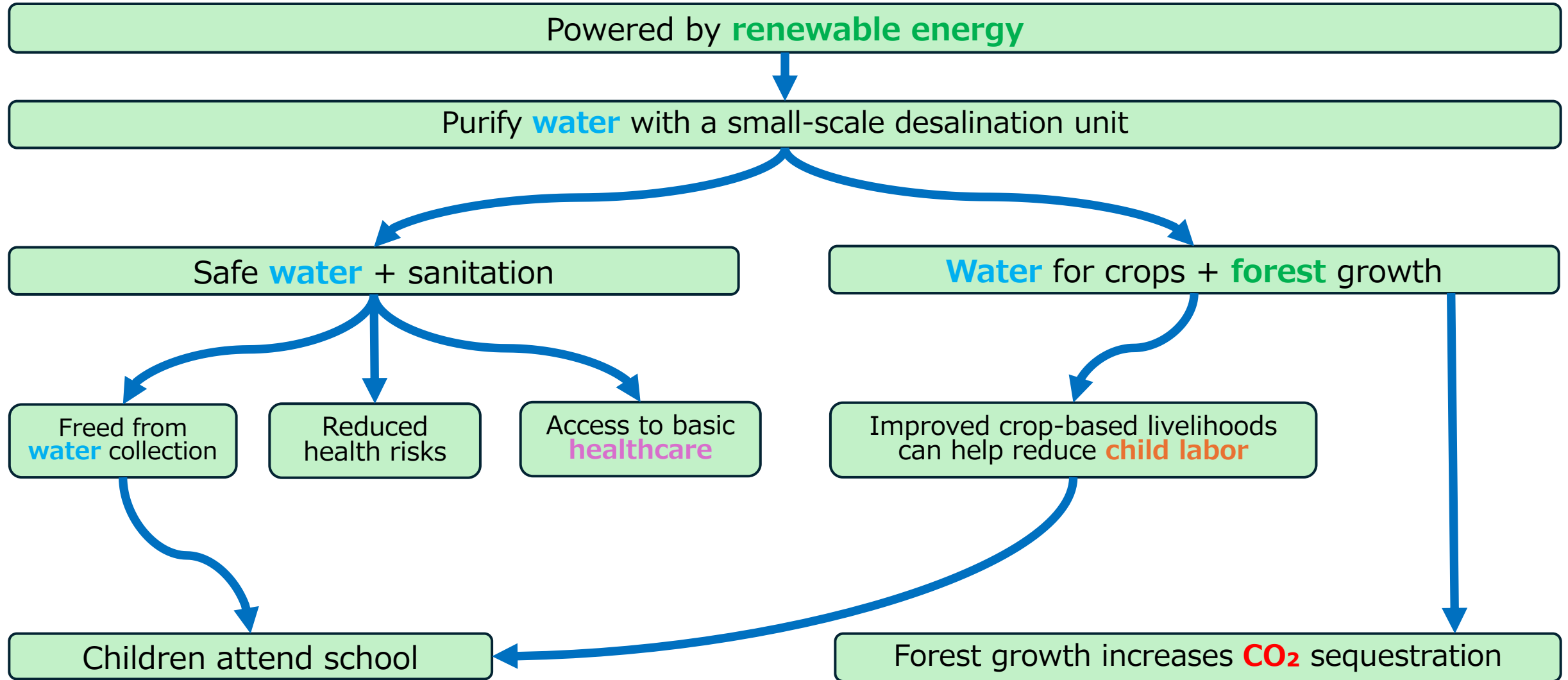
## **Agriculture & Environment**

- Food shortages
- Climate vulnerability
- **CO<sub>2</sub>** reduction

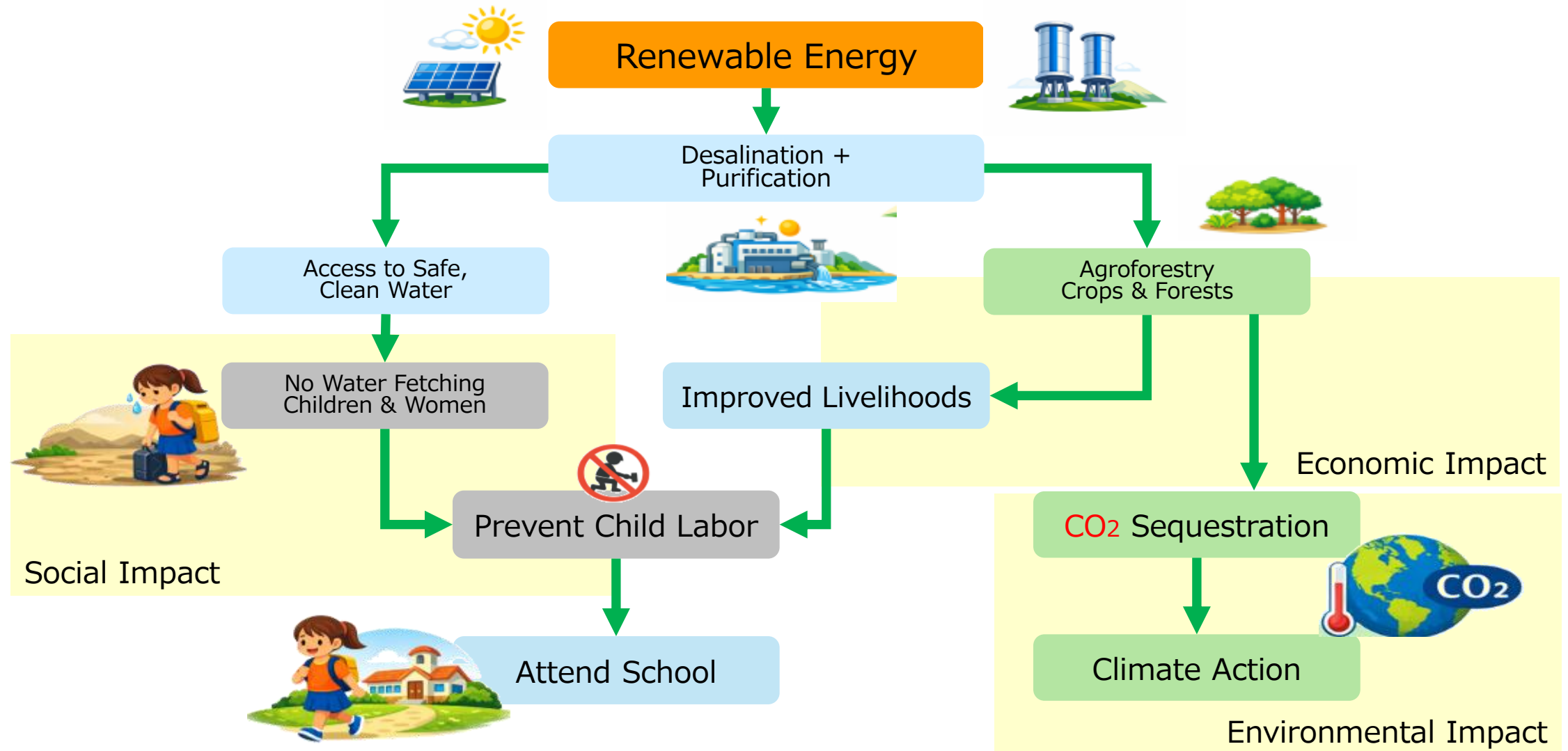
## 1-2. Key Technologies

| Category                  | Technology                                                                                | Impact Enabled                                                                                                                          |
|---------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Infrastructure            | <b>Small-scale seawater desalination equipment</b>                                        | Access to safe, sanitary <b>water</b><br>Safe toilets with lighting<br>Freedom from water collection<br>Water transport to inland areas |
| Health                    | Basic healthcare / Primary healthcare                                                     | Improved access to essential <b>health</b> services                                                                                     |
| Energy                    | <b>Renewable energy</b><br>Silicon photovoltaics<br>Perovskite photovoltaics<br>Biodiesel | Power source for desalination                                                                                                           |
| Agriculture & Environment | <b>Agroforestry</b>                                                                       | Livelihood improvement and local development through crops<br>Helps reduce <b>child labor</b> and support school attendance             |

## 1-2. SDGs: Overall View of a Simultaneous Approach to Multiple Issues



# 1-3. SDGs: Illustration of a Simultaneous Approach to Multiple Issues



# 1-3. SDGs: Illustration of a Simultaneous Approach to Multiple Issues

## (1) Access to Basic Healthcare

Improving access to essential healthcare services for children and families.



## (2) Safe Toilets and Nighttime Safety

Enabling girls and other children to safely use toilets at night.



## (3) Lighting for Nighttime Study

Enabling children to study safely at night with LED lighting.



# 1-4. Environmental Regeneration Effects of Agroforestry

- Climate Change Mitigation (Increasing CO<sub>2</sub> Sequestration)
  - Increase CO<sub>2</sub> sequestration through forest growth and agroforestry.
  - Contribute to climate change mitigation and help reduce climate-related risks.
- Regeneration of Regional Ecosystems
  - May contribute to changes in local water-cycle dynamics through increased forest cover.
  - Improve dryland climate adaptation by greening the Sahel region.

## **2. About Water**

## 2-1. Water Requirements (Drinking Water + Domestic Water)

- UNICEF Reference level

15 L/person/day                      of which drinking water is 3–5 L

[https://www.unicef.or.jp/library/pres\\_bn2007/pres\\_07\\_33.html](https://www.unicef.or.jp/library/pres_bn2007/pres_07_33.html)

- Assumption in This Model

30 L/person/day    Amount assuming improved living standards.

- Water required for a settlement of 1,000 people

$30 \text{ L} \times 1,000 \text{ people} = 30,000 \text{ L/day}$

## 2-2. Water Requirements (Crops + Forest Growth)

### Water required for crops and forests

Estimated water requirement per hectare for dryland agroforestry: **8,000–16,000 L/day/ha**

Based on CIFOR (Center for International Forestry Research) dryland studies and technical documents such as “Agroforestry and Water Management in Drylands.”

Reference: CIFOR Publications (<https://www.cifor.org/knowledge/publications/>)

Considering evaporation on extremely hot days, salt accumulation prevention, and climate-change risk (buffer), this model adopts **14,250 L/day/ha** as a design value.

When cultivated land is 20–40 ha, a large amount of water is required.

To avoid higher installation costs for small-scale desalination units, adopt the water-saving technologies on the next page.

## 2-3. Water-Saving Technologies

### Optimizing Water Retention for Crops and Forests

An approach to prevent excessive soil absorption and evaporation by “narrowing the outlet,” “covering the surface,” and “increasing water retention.”

- **Drip Irrigation**
- **Mulching (ground cover)**
- Use of water-retaining materials and soil conditioners
- Placement of windbreak forests (windbreaks)
- Subsurface irrigation (watering underground)

## 2-4. Water Requirements (1,000-person Settlement + 40 ha)

- Drinking & domestic water (1,000 people): 30,000 L/day
- Agroforestry water (40 ha): 570,000 L/day
  - \*Optimized by applying water-saving technologies (drip irrigation + mulching)

■ Total daily water requirement 600,000 L/day

### ■ Supply Plan

- Water storage tank capacity: 900,000 L

Always secure 1.5 days of supply; stable even during weather changes and maintenance

- Transport system: Supply each inland area from desalination hubs using biodiesel water trucks

[Supplement: Required water by farmland area]

1 ha: 14,250 L / 20 ha: 285,000 L / 30 ha: 427,500 L / 40 ha: 570,000 L

# **3. About Seawater Desalination Systems**

## 3-1. Small-Scale Seawater Desalination Unit



**Purifies seawater and  
contaminated (raw) water.**

- Desalination of seawater
- Purification of river and contaminated water through filtration systems
- Purification using RO (reverse osmosis) membranes
- Removes pathogens, viruses, and other contaminants
- Can remove microplastics
- Provides safe, treated water
- Stable supply through distributed deployment due to compact size
- Track record in overseas water and sanitation projects

## 3-1. Small-Scale Seawater Desalination Unit

### [Secondary Effects]

This unit uses RO reverse-osmosis membranes and removes fine particles in seawater with high efficiency.

The current process has demonstrated the potential to remove more than 99% of microplastics in the 1–5  $\mu\text{m}$  size range.

### [Future Development Potential]

By adding a dedicated module applying RO reverse-osmosis membrane technology, microplastic removal from seawater could be further enhanced.

This suggests potential for further investigation as an approach to addressing marine pollution.

### [Application to Inland Water Sources]

Although this unit is designed primarily as a small-scale seawater desalination system, it can also be used to purify freshwater sources such as river water, depending on water quality and pretreatment requirements.

## **4. About Renewable Energy**

## 4-1. About Renewable Energy

### ■ Silicon Photovoltaics

- Conventional solar power technology
- Based on the widely used 400 W panel.
- Use salt-damage-resistant, corrosion-resistant panels

### ■ Perovskite Photovoltaics

- A solar power technology expected to be deployed in the future
- Can generate power when applied to cylinders or glass surfaces
- Could become more cost-competitive than silicon photovoltaics in the future

## **5. Small-Scale Seawater Desalination and Required Number of Solar Panels**

## 5-1. Required Units (**Small-scale seawater desalination equipment**) and Solar Panels

|                                                                                  | Seawater                                     | Surface Water                            |
|----------------------------------------------------------------------------------|----------------------------------------------|------------------------------------------|
| Daily Water Demand                                                               | <b>600,000 L *</b>                           | <b>600,000 L *</b>                       |
| Solar Power Generation Hours                                                     | 7:00-17:00(10h)                              | 7:00-17:00(10h)                          |
| Water Production (10h/day)                                                       | 2300 L/10hr                                  | 4600 L/10hr                              |
| Number of Units Required                                                         | Approx. 260 Units                            | Approx. 130 Units                        |
| Operating Voltage                                                                | Single-phase 200V                            | Single-phase 200V                        |
| Power Consumption                                                                | 3,810 W(Unit+Pump)<br>3,360 W(Unit Only)     | 1,990 W(Unit+Pump)<br>1,540 W(Unit Only) |
| Required Number of Solar Panels (400 W)<br>(Upper: Rated, Lower: 70% Efficiency) | Approx. 2,480 panels<br>Approx. 3,540 panels | Approx. 650 panels<br>Approx. 930 panels |

\* Slide 13: 2-4: Water requirements for 1,000 people and 40 ha.

"260 units" is required due to the high pre-treatment load for seawater.

Figures are based on the current model (**Small-scale seawater desalination equipment**).

Appendix cost estimates adopt the 70% efficiency figures (Seawater: 3,540, Surface Water: 930).

## 5-2. Selection of Water Intake Sites and Water Supply

### ■ Selection Criteria for Water Intake Sites

- Physical requirements: Calm sea (seawater desalination) or rivers, lakes, aquifers
- Sustainability: Stable intake volume throughout the year  
(minimum: 600 m<sup>3</sup>/day)
- Social requirements: Sufficient consensus-building with local residents

### ■ Delivery & Logistics (areas where securing water sources is difficult)

- Securing delivery methods:
  - Current: Biodiesel water trucks
  - Future: EV water trucks equipped with sodium-ion batteries

## **6. Agroforestry: Crops and Trees**

## 6-1. Agroforestry Structure and Value

| Perspective          | Role of Agroforestry                                           | Economic & Environmental Value Created                                       |
|----------------------|----------------------------------------------------------------|------------------------------------------------------------------------------|
| Climate Resilience   | Trees shade intense sunlight and moderate microclimates        | Helps reduce the risk of crop loss during droughts (risk diversification)    |
| Soil Environment     | Leaf litter and roots add organic matter and nurture soil      | Reduces chemical fertilizer dependence and improves long-term soil fertility |
| Revenue Structure    | Multi-crop, multi-layer cultivation diversifies harvest timing | Stabilizes income year-round and breaks the poverty cycle<br>*1              |
| Regional Development | Adoption of the method triggers local ecosystem regeneration   | Improves working conditions and builds sustainable communities               |

\*1 Helps reduce child labor tied to household income.

# 6-2. Combination of Locally Suitable Crops and Trees

## ■ Crops/Trees to Grow and Their Roles

| Category                  | Role                                             | Key Examples                                                        |
|---------------------------|--------------------------------------------------|---------------------------------------------------------------------|
| Livelihood Improvement    | Cash income from harvest and sales               | Sweet potato, okra, cassava, vanilla, cacao, mango, papaya, avocado |
| Environmental Restoration | Soil improvement & CO <sub>2</sub> sequestration | Moringa, acacia, teak                                               |
| Fuel                      | Fuel for biodiesel water trucks                  | Moringa, jatropha, sunflower, castor bean, rapeseed                 |

Note: Crop and tree selection should respect local food culture and expert guidance.

# 6-3. Global Environmental Contribution of 10,000 Sites

40 ha × 10,000 sites could result in approximately 6 million tons of annual CO<sub>2</sub> sequestration.

Roughly comparable to the annual CO<sub>2</sub> emissions of Iceland.

\*Model case; results vary by vegetation density and climate conditions.

Average estimates below use a 1,000-person settlement model (20–40 ha), since carbon sequestration varies by crop/tree type.

| Cultivated Area<br>(per settlement) | Estimated Annual CO <sub>2</sub><br>Sequestration<br>(per site) | Number of<br>Projects | Estimated Total Annual CO <sub>2</sub><br>Sequestration |
|-------------------------------------|-----------------------------------------------------------------|-----------------------|---------------------------------------------------------|
| 20 ha                               | 200–400 tons                                                    | ×10,000               | 2–4 million tons                                        |
| 30 ha                               | 300–400 tons                                                    | ×10,000               | 3–4 million tons                                        |
| 40 ha                               | 400–800 tons                                                    | ×10,000               | 4–8 million tons                                        |

The 6 million tons figure is an illustrative model estimate.

## **7. Application Sites**

## 7-1. Target Settlements Requiring Deployment of This Integrated Model

- Africa: 54 countries

Africa has a population of 1.4 billion; over 400 million people are estimated to lack access to safe water.

- African settlements

300–1,500 people (most common)

2,000–5,000 people (medium-sized)

10,000+ people (treated as towns)

- Settlements with water issues

Hundreds of thousands of settlements in Africa face water-related issues.

Depending on the country, thousands to tens of thousands of settlements may be target sites.

- Estimated applicable areas

A preliminary conceptual estimate is around 500,000 settlements.

## 7-2. Target Regions Requiring Deployment of This Integrated Model

- Applicable to Southeast Asian regions with water issues  
Bangladesh, Cambodia, Laos, Myanmar, parts of Indonesia, parts of the Philippines
- Applicable to other regions as well  
Desert and semi-arid regions across the Northern Hemisphere  
Potentially deployable in ways that consider water resources and ecosystems.  
  
Northern China  
Around Mongolia  
Central Asia (southern Kazakhstan, Uzbekistan, Turkmenistan)  
Middle East (Jordan, northern Saudi Arabia, western Iraq)

## **8. Other Considerations**

## 8-1. Other Considerations: Do Not Harm

### ■ **Must not become a source of conflict**

Facilities built by this project must not become a flashpoint between the national government and anti-government armed groups.

Local residents' understanding is needed to avoid any sense of unfairness between project areas and non-project areas.

## 8-2. Other Considerations: Implementation Sites

- **Implementation sites (solar panels, small-scale seawater desalination, agroforestry)**

Land should be secured through state ownership or community-agreed land-use rights

- **Water allocation**

How to decide allocation of drinking/domestic water, agricultural water, and water for growing forests.

- **Salt production**

Concentrated brine discharged from small desalination/water purification units (when using seawater) can be used to produce salt.

Though limited, it can help improve local livelihoods and may cover maintenance costs such as filters.

## 8-3. Other Considerations: IT-Related

### ■ **Battery Selection: Avoid Conflict-Mineral Risks**

Adopt cobalt-free LFP (lithium iron phosphate) batteries to reduce conflict-mineral and child-labor risks

while maintaining safety in high-temperature environments.

Note: Lithium-ion batteries may contain conflict minerals such as cobalt, and harsh child labor at mining sites is a concern.

To reduce fossil-fuel dependence, avoid generators powered by fossil fuels.

If desalination must operate after sunset, use electricity from biodiesel generators.

Consider reducing the number of units; until large sodium-ion batteries become commercialized, purify as much water as possible during daylight hours and store it in tanks.

Note: Biodiesel is a renewable fuel, but its lifecycle greenhouse-gas emissions depend on the feedstock, production process, transport, and land-use impacts.

## 8-4. Other Considerations: Operations

### ■ **Build an IT-based remote monitoring system**

Monitor power generation, detect desalination/purification unit anomalies, and alert for filter replacement.

Because the units do not support remote monitoring, monitor equipment status with remote cameras.

Note: Saving camera footage and temporarily storing logs will be designed during implementation.

### ■ **Equipment maintenance**

Regular maintenance is required for desalination/purification units and solar power systems.

### ■ **Lightning rods**

Always install lightning rods to prevent equipment failure caused by lightning strikes.

## 8-5. Other Considerations — Risk Avoidance: Desert Locust Countermeasures (Short / Medium / Long Term)

### ■ Risks posed by desert locusts

- Can devastate crops and forests in a short time

- Gregarious swarms range from tens of millions to billions of individuals

- Travel more than 100 km per day

- The early stage of agroforestry is the most vulnerable

### ■ Short term: Leverage FAO's early warning system

- FAO Locust Watch: continuously monitor outbreak status, migration forecasts and alert levels.

### ■ Medium term: Eliminate at the larval (hopper) stage (carried out at village level)

- Larvae (hoppers) have no wings and only walk on the ground — this stage is the easiest to control

- Methods: herd into trenches and bury, barrier traps, small-scale spraying (conventional spraying, not ULV)

- Can be addressed at the village level → a potentially practical way to reduce the risk of major outbreaks

### ■ Long term

- Increasing forest cover

- Natural predators (birds, insects, small animals)

- Soil stabilizes and desertification is curbed → over the long term this reduces the very breeding grounds of desert locusts

## **9. What to Expect in the Future**

## 9-1. Increase in Vegetation Through Natural Rainfall

By deploying this model across multiple sites in suitable regions, increased vegetation and evapotranspiration could potentially contribute to changes in local water-cycle dynamics in some arid areas.

Under favorable regional climate and atmospheric conditions, changes in water-cycle dynamics may influence local rainfall patterns. In addition to planted forests, naturally regenerated vegetation could spread, potentially increasing CO<sub>2</sub> sequestration.

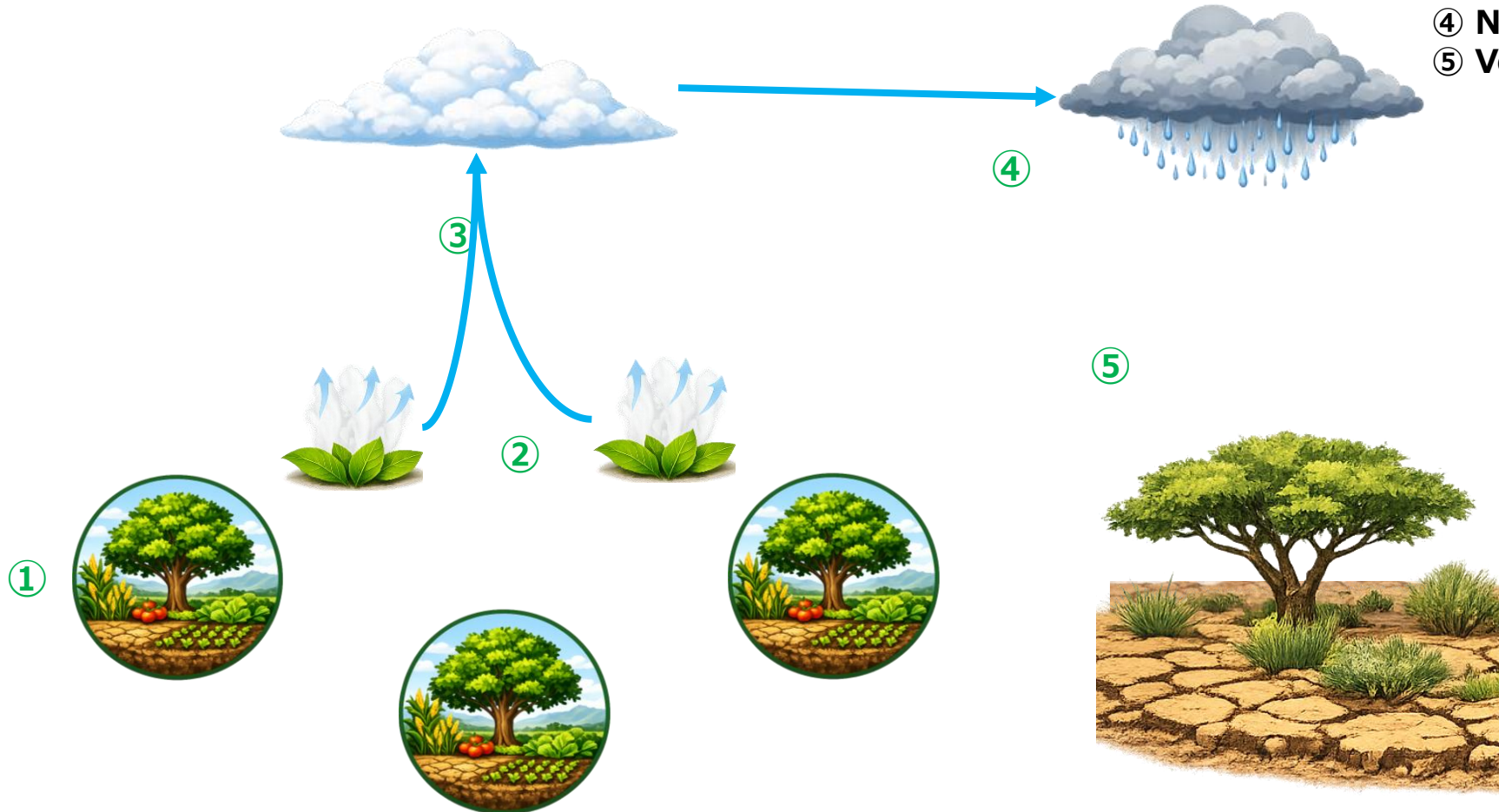
This could potentially contribute to a positive feedback loop that supports vegetation recovery and may help reduce desertification in some arid regions.

The specific implementation regions and forest areas should be determined by experts, but for example, in an arid area about the size of Narita Airport (approx. 1,000 ha), restoring about 100–300 ha as forest could potentially influence local water-cycle dynamics

Under favorable regional conditions, this could potentially contribute to changes in local rainfall patterns and support water-cycle recovery in surrounding areas.

## 9-2. Natural Rainfall Mechanism — Conceptual Image

- ① Deploy multiple agroforestry sites
- ② Transpiration from forests
- ③ Increased evapotranspiration  
→ atmospheric moisture and cloud formation
- ④ Natural rainfall
- ⑤ Vegetation establishes naturally



## 9-3. Crop Production + Forest Growth with Other Farming Methods

Where seasonal rainfall is sufficient

- **Half-moon farming**

- Rainwater collects effectively in the basins

- **Zai farming**

- Water and nutrients gather in the holes, sharply increasing germination rates

- **Stone lines**

- Water flow slows, helping soil remain instead of being washed away

- **FMNR (Farmer Managed Natural Regeneration)**

- A core technique for forest restoration, making forest regeneration easier

## 9-4. Reducing Hunger and Conflict

Through this model, water access, agriculture, and livelihoods may become more stable, potentially helping to **reduce hunger**.

Disputes over local resources and migration for livelihoods may decrease, potentially reducing the risk of “**competition-driven conflict**.”

This not only satisfies the basic principle of international cooperation, “Do No Harm (avoid causing harm)” but could also contribute to regional stability and peacebuilding.

# 10. Summary

# 10-1. Summary: Three Areas of Impact (1/3)

## 1. Water Security & Health:

Providing stable access to safe water and sanitation, reducing the burden of water collection, improving access to basic healthcare, and enabling educational opportunities.

These improvements can also contribute to reducing preventable maternal and newborn deaths.

## 2. Agroforestry & Ecosystem Regeneration:

Supporting economic growth, helping to reduce child labor associated with household poverty, and contributing to ecosystem regeneration.

## 3. Sustainable Community Building:

Creating infrastructure powered primarily by renewable energy, while reducing reliance on fossil fuels.

### **A Key Condition for Sustainable Implementation**

Fair access to water and land, equitable benefit-sharing, and transparent governance involving local communities are essential for sustainable implementation.

# 10-1. Summary: Three Areas of Impact (2/3)

## **Standardization of the 1,000-Person Unit Model**

### ■ **Unified Design:**

Standardized capacity, equipment, and renewable energy configuration for any region.

### ■ **Versatile Application:**

Fully compatible with both seawater and freshwater sources.

### ■ **High Scalability:**

Designed for broad implementation across diverse countries

# 10-1. Summary: Three Areas of Impact (3/3)

## Scaling Potential & Vision

### ■ Broad Reach:

Applicable to hundreds of thousands of communities across Africa and Asia.

### ■ Environmental Impact:

Potentially achieving approximately 6 million tons of annual CO<sub>2</sub> sequestration through 10,000 sites.

### ■ Social Impact:

Improving health and quality of life for children and families through simultaneous water, sanitation, healthcare, and education improvements.

### ■ Global Contribution:

Proposing a Japan-led international model to address global challenges.

"Targeting Broad Deployment by 2040"

## 10-2. Relationship with the SDGs

I want to improve access to safe water, sanitation, and basic healthcare for children and families.



Improving livelihoods through crops  
Escaping poverty through regional development



Access to basic healthcare  
Improved child and family health



Relieve women and girls of water-carrying duties  
Education → better jobs → prevention of child marriage



Use of renewable energy



CO<sub>2</sub> sequestration through tree planting and agroforestry



Securing food through crop harvests



Reduce the burden of water collection  
Higher incomes reduce child labor and enable school attendance



Securing safe, hygienic water  
Safe toilets with electric lighting



Reduce Child Labor



Forest growth  
Crops through drip irrigation and mulching

## 10-3. Additional Potential Impacts

| Issue                                    | Method                                                       | Effect                                                      | SDGs     |
|------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|----------|
| Nutrition / food insecurity              | Diverse crops and improved livelihoods                       | Improved child health and development; improved food access | SDGs 2,3 |
| Water collection burden                  | Access to safe and hygienic water                            | Women and children relieved from water-collection duties    | SDGs 5,6 |
| Maternal and newborn health              | Safe water and sanitation; access to basic healthcare        | Reduced risk of preventable maternal and newborn deaths     | SDGs 3,6 |
| Child labor                              | Safe water, agroforestry, improved livelihoods through crops | Reduced child labor undertaken to support household income  | SDGs 8   |
| Child marriage                           | Improved livelihoods and restored educational opportunities  | Reduced risk of child marriage                              | SDGs 5   |
| Nighttime safety for girls               | Renewable-energy lighting, security cameras and patrols      | Girls can safely use toilets at night                       | SDGs 5,6 |
| Lack of nighttime learning opportunities | Renewable energy + lighting                                  | Opportunities to study at night                             | SDGs 4,7 |
| Energy shortage                          | Renewable energy                                             | Improved access to lighting and water treatment             | SDGs 7   |

## 10-4. Toward a Sustainable Cycle of Support

No single project can reach everyone in need. It is difficult to reach everyone through ODA and donations alone. Where this approach cannot reach, I hope NGO support can continue to provide assistance to those who need it most.

As more communities improve their livelihoods through this approach, I hope that NGO activities and other forms of external support can reach people and communities that still need assistance.

Looking further ahead, I believe an even more positive cycle could emerge if some of those whose lives have improved through support go on to support NGOs and others in need themselves.

## 10-5. Small-Start and Wider Application

This approach can start on a small scale.

This proposal uses a community of 1,000 people and 40 hectares of agroforestry land as a model. However, the scale can be adjusted flexibly according to the population and available agricultural land. This scalability is one of the strengths of the approach.

I learned about initiatives such as the Sahel Forest Project (SFP) and the Great Green Wall (GGW) after developing this proposal. I hope that this approach can be implemented not only in specific regions, but across many regions.

I would also be pleased if this approach could contribute, even partially, to securing water sources and supporting other projects such as the Great Green Wall and projects being implemented in Asia.

## 10-6. Community Development and Wider Benefits

- As livelihoods improve, communities can gradually expand solar power capacity to support further development.
- Biodiesel-powered vehicles can facilitate the transportation and distribution of agricultural products.
- As food, clothing, housing, and other basic needs become more stable, people can devote more time and resources to education, work, and other community activities.
- Where capacity allows, communities can share water and other resources with neighboring communities facing similar challenges.

## 10-7. Contribution to Child and Women's Well-being

This integrated approach could contribute to improving the health and well-being of children and women, while also complementing existing activities by UNICEF and child-support NGOs, including child sponsorship programs.

This proposal was created for **children of the future.**

I want to improve the health and well-being of children and families.

I want to reduce the number of children who must fetch water,  
work to support their families, and give up school.

I want to help reduce atmospheric **CO<sub>2</sub>** and help reduce the risks children  
face from heat waves and droughts.

I want to create a future where, as adults, they can help build hope  
for the next generation.

That is the spirit behind this proposal.

**Thank you for your consideration.**

# Open Knowledge Statement

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I am entrusting this to a promising young person.

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This proposal receives technical cooperation from Waqua Co., Ltd. for compact seawater desalination equipment (MYZ-E series).

\*This document does not cover detailed specifications related to corporate confidential information.

## ■ Revision History

[illegible]